

Question Paper

Exam Date & Time: 29-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026
SUBJECT: ECE 3122/ECE_3122 MICROPROCESSORS
(SPL: ELECTRONICS & COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

Missing data may be suitably assumed.

- 1A) Explain the concept of banked registers in the ARM architecture. How do banked registers facilitate efficient context switching between different operating states (modes) of the ARM processor? Illustrate your answer with examples of registers banked for different modes. (4)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) Analyse the key characteristics of RISC architecture. Compare it with CISC architecture, highlighting at least three major differences. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 1C) Explain the purpose of the 'Align' directive in an assembler? How does it ensure proper memory alignment? Elucidate with the help of an example. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2A) Describe the different types of exceptions in the ARM7 processor and explain how each is handled. If multiple exceptions occur simultaneously, how does the processor resolve this situation? (4)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2B) Discuss different methods of parameter passing to subroutines in ARM7 assembly language programming, and illustrate each method with examples. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 2C) With a neat block diagram, discuss the datapath activity when ARM executes the machine code 0xEAFFFFFF present at the memory address: 0x00000018 (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3A) Pre-conditions: Assume that the memory of ARM is word-aligned and data in any 32-bit memory location is the 1's complement of its word-aligned address, the value of R0 is 0x5000, and the value of R1 is 0x20;
With the given pre-conditions, compute the values of R0 and R1 due to the execution of below instructions:
i) LDR R1, [R0, #4]
ii) LDR R1, [R0, R1]! (4)

- iii) LDR R1, [R0, R1, LSL #4]!
- iv) LDR R1, [R0, R1, LSR #2]
- v) LDR R1, [R0], #4
- vi) LDR R1, [R0], R1
- vii) LDR R1, [R0], R1, LSL #4
- viii) LDR R1, [R0], R1, ROR #2

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 3B) Explain the role of the Barrel Shifter in the ARM data processing model and describe different types of operand shifts and rotations supported by the barrel shifter during instruction execution. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 3C) Write the shortest possible ARM assembly program to implement the following mathematical expression: (3)

$$Y = \sum_{n=0}^2 a * X[n]$$

Assume that the variables a , $X[0]$, $X[1]$ and $X[2]$ can hold 32-bit data, whereas variable Y can hold 64-bit data.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 4A) Write the shortest possible ARM assembly program to convert a 32-bit gray code to a binary number. (4)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 4B) In ARM, what would be the target address space that the branch instruction "B" can cover? Give the reasons behind it and explain the possibilities to branch to the entire 4GB address space, which is supported by the ARM. (3)

Course Outcome: "CO1" / Cognitive Level: "Remember"

- 4C) Pre-conditions: R1=0x1, R2=0x2, and R13=0x80000008 (3)

Memory address	Data
0x80000010	0x00002004
0x8000000C	0x00002003
0x80000008	0x00002002
0x80000004	0x00002001
0x80000000	0x00002000

What would be the value of stack pointer and the values in memory locations addressed from 0x80000000 to 0x80000010 due to the execution of below ARM instructions:

- i) STMFA R13!, {R1,R2}
- ii) STMED R13!, {R1,R2}
- iii) LDMFD R13!, {R1,R2}
- iv) LDMEA R13!, {R1,R2}
- v) STMFD R13!, {R1,R2}
- vi) LDMFA R13!, {R1,R2}

Course Outcome: "CO1" / Cognitive Level: "Remember"

- 5A) Interface two switches (SW0 and SW1) and eight LEDs to the LPC2148 microcontroller using port lines P0.0-P0.1 for the switches and P0.2-P0.8 for the LEDs. Write an embedded C program to (4)

perform the following:

- When SW0 is pressed, implement a Johnson counter.
- Otherwise, blink alternate LEDs simultaneously with a suitable delay.

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- 5B) Draw the schematic of interfacing a stepper motor to LPC2148, and explain the configuration of all GPIO registers that are necessary to run the motor. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5C) For the generic ARM chip, compute the following: (3)
- The amount of memory (in bytes) given to the address range of 0x40000000 - 0x400FFFFF for the peripheral bit-addressable region.
 - The amount of memory (in bytes) given to the address range of 0x42000000 - 0x43FFFFFF for alias addresses of bit-addressible SRAM.
 - The bit-alias address of bit 5 of 0x40000008

Course Outcome: "CO3" / Cognitive Level: "Application"

-----End-----

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ECE_3122:MICROPROCESSORS

#	Course Outcome	Description	Marks
1	1	CO 1	20
2	2	CO 2	7
3	3	CO 3	9
4	4	CO 4	10
5	5	CO 5	4
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	ECE_3122:MICROPROCESSORS

#	Level	No of Questions	Marks (%)
1	Remember	2	6(12%)
2	Understand	4	14(28%)
3	Application	4	12(24%)
4	Analysis	4	14(28%)
5	Create	1	4(8%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)